GraphPapers

`v1.3`

A free, single-file alternative to Connected Papers. Enter a paper, get a force-directed graph of the ~40 works most related to it, then export a ready-made prompt that turns that graph into a narrative history — in your own LLM, at no cost.

No install. No account. No API key. No backend. One HTML file.

Running it

Open it at **https://oddthumb.github.io/GraphPapers/**, or download `index.html` and double-click it. Either way works, and both are the whole procedure.

Everything the app needs to run is inside that file, including the d3 graph library. The only thing it fetches from the network is the paper data itself, live from OpenAlex — a free, open catalogue of 250M+ works that needs no key and imposes no cost.

Sharing it means sending a link, or one file. Either way the recipient is working immediately.

What it does

Similarity graph

Enter a title, DOI, or OpenAlex ID. The app pulls the seed paper's references and the works citing it, then scores every candidate against the seed using the same two measures Connected Papers uses:

- **Bibliographic coupling** — two papers that cite many of the same works are probably about the same thing.
- **Co-citation** — two papers that are frequently cited together are probably about the same thing.

Both measures are standard bibliometrics, not anyone's proprietary method — bibliographic coupling is Kessler (1963), co-citation is Small (1973).

The 40 highest-scoring papers become the graph. Each node connects to its three nearest neighbours, so clusters emerge on their own.

An edge means similarity, not citation. Two papers are joined because they cite the same works or are cited together — they need not cite each other at all. Where a real citation does exist between two joined papers, the edge carries an arrowhead pointing at the older work. To see citations on their own terms, tick **Citation links**: that overlays every real citation between the papers on screen, on top of the similarity layout, without moving anything.

ENCODING	MEANING
Node size	Citation count
Node colour	Publication year — silver (older) to blue-slate (newer)
Coral ring	The seed paper
Edge weight	Similarity between the two papers

Everything that changes the graph lives in one panel at its top-left; everything that only reads from it — **Story prompt**, **Export citations** — stays in the tab bar. The **i** button at the top-right opens the full legend.

Drag nodes, scroll to zoom, and use the **Spacing** slider to loosen or tighten the layout. **Reshuffle** throws the nodes to new starting positions and lets the layout settle again — a force layout falls into whichever arrangement its start positions lead to, so a tangled graph often untangles on the second try. **Timeline** stacks the papers by year instead, oldest at the top, so citation arrows all point upward into the past; each year that actually occurs gets an equal band, so a single old reference cannot squash the recent decade into a sliver. Three colour palettes sit in the header.

Click a node to pin its neighbourhood; click empty canvas to release it. Click any node to see its abstract, authors, and DOI — and to rebuild the whole graph around it.

Prior and derivative works

Two tabs beside the graph list what the seed paper cites and what cites it, ranked by citation count.

Citation export

Export citations writes the papers out as BibTeX, RIS, APA, MLA, or Chicago — one paper, everything the filters currently show, or all 40. Copy it, or download a `.bib` / `.ris` file and drag it straight into Zotero, EndNote, or Overleaf.

Author names come from OpenAlex as single strings, so the family name is taken as everything after the last space. That is right for most Western names and wrong for some; check the names before submitting.

Filtering

In the same left-hand panel: keyword (title and authors), year range, minimum citations, open access, and journal impact. The journal filter uses OpenAlex's 2-year mean citedness, ranked among the journals present in the current graph — it is **not** a JCR quartile, which is not open data. Filtering hides papers without moving the layout, and resets whenever you build a new graph.

LLM story prompt

The **Story prompt** button assembles a prompt describing the graph — every paper grouped as prior, later, or related, with its year, first author, citation count, and distance from the centre in graph hops.

Copy it into ChatGPT, Claude, Gemini, or anything else, and you get a written history of the research line: what problem the centre paper solved, what led to it, and what it produced.

DEPTH	PAPERS	USE IT FOR
1	~10	A tight story about the paper's immediate neighbourhood
2	~25	The default — enough context to see a lineage
3	~37	The whole field around the paper
All	39	Everything in the graph

Output language is switchable between English and Korean.

The app builds this text itself — it never calls an LLM, so this step costs nothing, needs no key, and works instantly. Which model reads it is entirely your choice.

Data freshness

There is no local cache and no database. Every lookup goes straight to OpenAlex, so what you see is what OpenAlex holds at that moment. The bottom-right corner always shows the time of the current fetch and the date OpenAlex last updated the seed record, so you know exactly what you are looking at.

The cost of that guarantee is that **the app needs an internet connection**. Without one it opens and renders, but cannot fetch papers, and says so.

Requirements

A modern browser. That is all.

Google Fonts is the only other external reference; if it is unreachable the app falls back to system fonts and works normally.

Credits

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Ideas, bug reports, and pull requests are welcome; contributors are credited in the app, under the Acknowledgements link in the bottom-right corner.

Paper data from OpenAlex. Graph layout by d3.

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